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Pyrotechnic driving device

Abstract

A driving device, comprising a hand-held housing with a piston element arranged therein for the transmission of energy to a fastening element to be driven in a driving direction, a propellant charge, and a combustion chamber, which is arranged between the propellant charge and the piston element and extends around a central axis, and an actuator, by which the energy to be transmitted from the propellant charge to the piston element can be varied in such a way that the energy can be set, wherein a venting channel connected to the combustion chamber can be exposed by means of a movable slide of the actuator, wherein the venting channel opens out into the combustion chamber by a vent opening, which can be partially or completely covered by the slide, and wherein a cross-sectional area of the vent opening increases in the driving direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

- (1) This patent application is the U.S. National Stage of International Patent Application No. PCT/EP2022/056854, filed Mar. 16, 2022, which claims the benefit of European Patent Application No. 21165497.5, filed Mar. 29, 2021, which are each incorporated by reference.
- (2) The invention relates to a driving device.
- (3) The prior art discloses hand-held driving devices with propellant charges in which the combustion gases resulting after the ignition of a pyrotechnic charge expand in a combustion chamber. This has the effect that a piston is accelerated as an energy transmission means and drives a fastening element into a workpiece. Combustion residues can cause the combustion chamber to become contaminated.
- (4) U.S. Pat. No. 6,321,968 B1 describes a driving device with a propellant charge in which the combustion chamber is separated into an upper subchamber and a lower subchamber by means of an orifice plate. The driving device can be adjusted in the volume of its dead space in order to adjustably vary the driving energy of the device. For this purpose, a valve-like slide can be adjusted in a direction perpendicular to a driving axis. In this case, the combustion chamber also has a dead space in the closed position of the slide, the dead space being formed as a clearance in a side wall of the combustion chamber.
- (5) The object of the invention is to provide a driving device which allows driving energy to be easily set, if need be over as wide a range as possible, with a given propellant charge.
- (6) According to one aspect, a driving device comprises a hand-held housing with a piston element arranged therein for the transmission of energy to a fastening element to be driven in a driving direction, a propellant charge, in particular an exchangeable propellant charge, and a combustion chamber, which is arranged between the propellant charge and the piston element and extends around a central axis, and an actuator, by means of which the energy to be transmitted from the propellant charge to the piston element can be varied in such a way that it can be set, wherein a venting channel connected to the combustion chamber can be exposed by means of a movable slide of the actuator, wherein the venting channel opens out into the combustion chamber by a vent opening, which can be partially or completely covered by the slide, and wherein a cross-sectional area of the vent opening increases in the driving direction. As a result, during a displacement of the slide, at first only a very narrow portion of the vent opening and later a wider portion of the vent opening is exposed. This allows a largely linear dependence of the set energy to be transmitted to the piston element on the displacement distance of the slide. Preferably, the vent opening is arranged in a cylindrical, particularly preferably circular-cylindrical, portion of the combustion chamber.
- (7) An advantageous embodiment is characterized in that the driving device has a number of venting channels, which respectively open out into the combustion chamber by a vent opening and have in each case a cross-sectional area which increases in the driving direction.
- (8) According to a further aspect, a driving device comprises a hand-held housing with a piston element arranged therein for the transmission of energy to a fastening element to be driven in a driving direction, a propellant charge, in particular an exchangeable propellant charge, and a combustion chamber, which is arranged between the propellant charge and the piston element and extends around a central axis, and an actuator, by means of which the energy to be transmitted from the propellant charge to the piston element can be varied in such a way that it can be set, wherein the combustion chamber has an internal thread and a preferably cylindrical guiding portion, wherein the actuator is arranged in the combustion chamber and has an external thread in engagement with the internal thread and a bearing for mounting the actuator in the guiding portion, wherein a thread gap spacing between the external thread and the internal thread is greater than a bearing gap spacing between the bearing and the guiding portion. As a result, forces and torques which act on the actuator during the combustion in the combustion chamber are transmitted via the

bearing to the guiding portion, so that the external thread and the internal thread are relieved. Combustion residues caused by the combustion in the combustion chamber may contribute less to jamming of the actuator, so that the actuator remains more easily adjustable. Preferably, the thread gap spacing is at least twice, particularly preferably at least three times, as great as the bearing gap spacing.

(9) An advantageous embodiment is characterized in that the combustion chamber has a preferably cylindrical further guiding portion, wherein the actuator has a further bearing for mounting the actuator in the further guiding portion, wherein a thread gap spacing between the external thread and the internal thread is greater than a further bearing gap spacing between the further bearing and the further guiding portion. Preferably, the external thread is arranged between the bearing and the further bearing along the central axis. Likewise preferably, the further bearing gap spacing is substantially the same size as the bearing gap spacing.

(10) An advantageous embodiment is characterized in that the internal thread and/or the external thread does not have any thread runout.

(11) An advantageous embodiment is characterized in that the internal thread and/or the external thread has one or more transporting grooves, which in particular run parallel to the driving direction. A further advantageous embodiment is characterized in that the actuator has one or more transporting bores, which in particular run parallel to the driving direction.

(12) An advantageous embodiment is characterized in that a venting channel connected to the combustion chamber can be exposed by means of a movable slide of the actuator. A further advantageous embodiment is characterized in that a starting position of the piston element in relation to the combustion chamber can be set by means of the actuator.

(13) An advantageous embodiment is characterized in that the actuator comprises a threaded sleeve, which preferably has a cylindrical inner wall.

(14) Generally preferably, the piston element is guided in the slide, at least over a first part of its movement. Depending on the detailed design, variable exposure of the venting channel or number of venting channels can take place by an adjustment of the slide in an axial direction and/or in the circumferential direction.

(15) A venting channel in the sense of the invention is any space that can be selectively added to the volume of the combustion chamber by the setting element, in order to vary driving energy in a defined manner by way of the additional expansion space. Preferably, but not necessarily, the venting channel may in this case be connected to an external space. As an alternative or in addition to this, the venting channel may also be a volume of dead space that is not connected to an external space.

(16) In the sense of the invention, driving energy is understood as meaning the energy transmitted to a given fastening element with a given propellant charge. If these boundary conditions are specified, it is made possible by the actuator to vary the resultant driving energy of the fastening element in such a way that it can be set.

(17) In the sense of the invention, a piston element is any means to which kinetic energy is imparted by the ignition of the charge, with the kinetic energy ultimately being transmitted to the fastening element. The piston element often takes the form of a piston, in particular a cylindrical piston. The piston head may be provided with clearances or other structures which further promote turbulence and uniform expansion of the combustion gases.

(18) A central axis in the sense of the invention is an axis which is at least parallel to the movement of the fastening element and passes through a center of the combustion chamber. Preferably, the central axis passes both through the center of the combustion chamber and through a center of the fastening element.

(19) A fastening element in the sense of the invention is generally any anchorage that can be driven in, such as for example a nail, bolt or screw.

(20) Generally advantageously, the slide has a preferably cylindrical inner wall which is formed as

part of the combustion chamber. In the case of such an arrangement, the slide expediently also serves for guiding the piston element in a first portion of the movement of the piston.

(21) To achieve easy and intuitive adjustment of the driving energy, the actuator has an operating part which is pivotable about the central axis. The operating part may be any suitable means for manual adjustment, such as for instance a rotatable sleeve as a particularly preferred variant, a pivotable knob or the like. The pivotability of the operating part about the central axis has the effect of achieving easy adjustment and at the same time effective visual monitoring of the value that is set. Such an arrangement also allows easy adjustment even under unfavorable conditions, such as for example when wearing work gloves.

(22) Pivoting of the operating part about the central axis means in this case a deflection of the operating part from a previous position and oriented substantially perpendicularly to the axis. A line of moment or trajectory of the operating part has in this case a radius of curvature which is preferably not less than a distance of the operating part from the central axis. Preferably, but not necessarily, the pivoting is a rotation about the central axis. In this case, the operating part and the slide are preferably connected for conjoint rotation, so that turning of the operating part at the same time brings about turning and adjustment of the slide for regulating the driving energy.

(23) It is generally advantageously provided that the slide has a collar which runs around the central axis and engages in an overlapping manner in a recess of a combustion chamber housing, wherein at least part of the venting channel is formed as an axially extending clearance between the collar and the combustion chamber housing. In this way, as a result of the overlap, a tight seal can be achieved, even with respect to great gas pressures. Moreover, a connection between the venting channel and the combustion chamber may already be exposed directly at a charge-side end of the combustion chamber in such a way that it can be set, which allows adjustability over a great range of the driving energy.

Description

(1) Further features and advantages of the invention are apparent from the exemplary embodiment and from the dependent claims. A preferred exemplary embodiment of the invention is described below and explained in more detail with reference to the appended drawings.

(2) FIG. 1 shows a spatial overall view of a driving device according to the invention.

(3) FIG. 2 shows a longitudinal section of a combustion chamber.

(4) FIG. 3 shows a longitudinal section of a combustion chamber with an actuator in a position for low driving energy.

(5) FIG. 4 shows a longitudinal section of a combustion chamber with an actuator in a position for high driving energy.

(6) FIG. 5 shows a detail of the longitudinal section in FIG. 4.

(7) FIG. 6 shows a spatial view of a detail of an actuator.

(8) Shown in FIG. 1 is a driving device. The driving device comprises a hand-held housing 1, in which a piston element in the form of a piston is held. A surface of the piston delimits a combustion chamber, in which the combustion gases of a pyrotechnic charge expand in order to accelerate the piston.

(9) The piston to which kinetic energy is imparted acts with an end tappet on a fastening element (not shown), which is thereby driven into a workpiece (on the right in FIG. 1). The fastening element may be held in particular in a module or magazine (not shown), which is exchangeably attached in a forward holding region 1a of the driving device 1.

(10) The charge is in the present case held in a sheet-metal cartridge 4a. The cartridge 4a has an impact detonator and, before ignition, is inserted into a cartridge bearing by way of a corresponding charging mechanism, in the present case by means of a magazine strip 4. The cartridge 4a and the

cartridge bearing are then arranged rotationally symmetrically around a central axis A. The central axis A is in the present examples at the same time a central axis of the combustion chamber and of the piston element.

(11) Shown in FIG. 2 is a combustion chamber **20**, which is formed by a combustion chamber housing **21**. The combustion chamber **20** is arranged between a circular opening **22** of the cartridge bearing **23** and the surface of the piston (not shown). At the time of the combustion, the cartridge lies against the cartridge bearing and is enclosed there. In the combustion chamber housing **21** there is a multi-start internal thread **24** in the region of the combustion chamber **20**. Connected to the combustion chamber **20** are a number of venting channels **25**, in particular two or three, which in each case open out with a vent opening **26** into the combustion chamber **20**. A cross-sectional area of each vent opening **26** increases in the driving direction **27**. The vent openings **26** are arranged in a circular-cylindrical portion of the combustion chamber **20**. Furthermore, the combustion chamber **20** has a circular-cylindrical guiding portion **33** and a further guiding portion **32**. The internal thread **24** is arranged between the guiding portion **33** and the further guiding portion **32** along the central axis A.

(12) In FIGS. 3, 4 and 5, the combustion chamber **20** is shown with an actuator **28** arranged therein. The actuator **28** comprises a threaded sleeve with a cylindrical inner wall and also a piston guide **28a**. By means of the actuator **28**, the energy to be transmitted from the propellant charge to a piston element **34** is variable in such a way that it can be set, by the venting channels **25** being gradually exposed by means of the actuator **28** and additionally or alternatively by a starting position of the piston element **24** in relation to the combustion chamber **20** being set by means of the actuator **28**. The fact that the cross-sectional areas of the vent openings **26** increase in the driving direction **27** means that, during a displacement of the actuator **28**, at first only a very narrow portion of the vent openings **26** and later a wider portion of the vent openings **26** is exposed in each case. As a result, the energy to be transmitted to the piston element **34** has a linear relationship with the displacement distance of the actuator **28**.

(13) The actuator **28** has an external thread **29** in engagement with the internal thread **24** and a bearing **30** for mounting the actuator **28** in the guiding portion **33** and also a further bearing **31** for mounting the actuator **28** in the further guiding portion **32**. The external thread **29** is arranged between the bearing **30** and the further bearing **31** along the central axis A. A thread gap spacing between the external thread **29** and the internal thread **24** is approximately four times as great as a bearing gap spacing between the bearing **30** and the guiding portion **33** and/or a bearing gap spacing between the further bearing **31** and the further guiding portion **32**. As a result, the external thread **29** and the internal thread **24** are subjected to less loading by forces and torques occurring during the combustion.

(14) A flank angle between a front side **35** of the external thread **29** and/or the internal thread **24** and the central axis A is between 0° and 60°, preferably between 30° and 45°. A transition from the front side **35** to a thread base of the external thread **29** or to a thread back of the internal thread **24** is advantageously provided with a tree-root fillet.

(15) In FIG. 6, an actuator **60** is depicted in a spatial view of a detail. The actuator **60** comprises a piston guide **61** and a slide **62**, which is axially displaceable in relation to the piston guide **61**, but during rotation of the piston guide **61** about a central axis A' is turned with it. For this purpose, the piston guide **61** has two claws **63**, which engage in two corresponding recesses **64** in the slide **62** for conjoint rotation but axially displaceably. In exemplary embodiments that are not shown, the piston guide is coupled with the slide by means of one, three, four, five or more claws.

Advantageously, all of the claws come into engagement with the respective recesses at the same time, so that lateral forces on the slide are reduced. This simultaneous engagement is improved by the claws being designed as bending springs.

(16) The piston guide **61** is for its part connected for conjoint rotation to an operating part **10** (FIG. 1) for conjoint rotation, so that the piston guide **61** at the same time forms a mechanical connection

between the operating part **10** and the slide **62**. The operating part **10** forms together with the piston guide **61** and the slide **62** the actuator **60** for varying the driving energy of the driving device.

(17) The slide **62** has an external thread **65**, which is formed so as to complement an internal thread of a combustion chamber housing (not shown), for example the internal thread **24** (FIG. 2). The external thread **65** does not have a thread runout, but instead ends abruptly with a scraping edge **66**, which contributes to cleaning the combustion chamber of combustion residues when circumstances require. In order to allow or assist such combustion residues to be transported away, the external thread **65** has a number of transporting grooves **67** and also a number of transporting bores **68**, which in each case run substantially parallel to the driving device. As a result, the actuator **60** remains easily adjustable when circumstances require.

(18) The adjustment of the driving energy works for example as follows:

(19) The slide **62** is screwed with its external thread **61** into the internal thread of the combustion chamber housing. Turning of the operating part, and consequently of the piston guide **61** and the slide **62**, about the central axis A accordingly brings about a forcibly controlled axial shift of the slide **62**. In preparation for a driving operation, the desired driving energy is set as an energy stage marked on the operating part by turning the operating part. This leads by way of the forced control described above to a corresponding axial positioning of the slide **62** in relation to the combustion chamber housing. As a result, any venting channels there may be are partially exposed over the through-opening (low driving energy, FIG. 3) or closed (maximum driving energy, FIG. 4).

(20) The invention has been described with reference to a number of exemplary embodiments of a setting tool. It goes without saying that all the features of the individual exemplary embodiments can also be realized in a single device in any desired combination, as long as they are not mutually contradictory. It should also be noted that the invention is also suitable for other applications.

Claims

1. A driving device, comprising a hand-held housing with a piston element arranged therein for the transmission of energy to a fastening element to be driven in a driving direction; a propellant charge; a combustion chamber, which is arranged between the propellant charge and the piston element and extends around a central axis; and, an actuator, by which the energy to be transmitted from the propellant charge to the piston element can be varied in such a way that the energy can be set, wherein the driving device has a number of venting channels connected to the combustion chamber that can be exposed by a movable slide of the actuator, wherein the number of venting channels respectively open out into the combustion chamber by a vent opening, which can be partially or completely covered by the slide, and wherein the number of venting channels have in each case a cross-sectional area of the vent opening which increases in the driving direction.
 2. The driving device as claimed in claim 1, wherein the vent opening is arranged in a cylindrical portion of the combustion chamber.
 3. The driving device of claim 2, wherein the vent opening is arranged in a circular-cylindrical portion of the combustion chamber.
 4. The driving device of claim 1, wherein the propellant charge is an exchangeable propellant charge.
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